

Q1

Given:

$$T(n) = T(n/2) + T(2n/5) + 7n$$

$$T(0) = 1$$

Determine the asymptotic complexity of $T(n)$.

Solution

Using the Akra-Bazzi theorem:

Find p such that:

$$(1/2)^p + (2/5)^p = 1$$

For $p = 1$:

$$1/2 + 2/5 = 9/10 < 1$$

Since the sum is less than 1, the linear term $7n$ dominates.

Answer:

$$T(n) = \Theta(n)$$

Q2

Given:

$$T(n) = aT(n/b) + f(n)$$

For constants $a \geq 1$ and $b > 1$.

When will:

$$T(n) = \Theta(n \log n)$$

hold?

Solution

From Master Theorem:

$T(n) = \Theta(n \log n)$ occurs when

- $\log_b(a) = 1$
- $f(n) = \Theta(n)$

Thus:

Answer:

$a = b$ and $f(n) = \Theta(n)$

Q3

An algorithm checks whether an array of size N is sorted by making a single pass and comparing adjacent elements.

Solution

Number of comparisons:

$(N-1)$

Answer:

$\Theta(n)$

Q4

Evaluate:

$$1^3 + 2^3 + 3^3 + \dots + n^3$$

Formula:

$$1^3 + 2^3 + \dots + n^3$$

$$[n(n+1)/2]^2$$

=
 $\Theta(n^4)$

Answer:

$\Theta(n^4)$

Q5

```
int fun1(int n)
{
    int i,j,k,p,q=0;

    for(i=1;i<=n;++i)
    {
        p=0;

        for(j=n;j>=1;j/=2)
            ++p;

        for(k=1;k<=p;k*=2)
            ++q;
    }

    return q;
}
```

Solution

Outer loop:
 $\Theta(n)$

Middle loop:
 $\Theta(\log n)$

Thus $p = \Theta(\log n)$

Last loop:
 $\Theta(\log \log n)$

Total:

$\Theta(n \times \log \log n)$

Answer:

$\Theta(n \log \log n)$

Q6

```
for(int i=n;i>0;i/=2)
{
    for(int j=0;j<i;j++)
    {
    }
}
```

Solution

Iterations:

$n + n/2 + n/4 + \dots$

Geometric series:

$= \Theta(n)$

Answer:

$\Theta(n)$

Q7

```
for(int i=1;i<=n;i++)
{
    for(int j=1;j<=i;j++)
    {
    }
}
```

Solution

$1+2+3+\dots+n$

$$= \frac{n(n+1)}{2}$$

Answer:

$$\Theta(n^2)$$

Q8

```
void fun(int n)
```

```
{  
    if(n<=1)  
        return;
```

```
    fun(n/2);
```

```
    fun(n/2);
```

```
}
```

Recurrence:

$$T(n)=2T(n/2)+1$$

Using Master theorem:

Answer:

$$\Theta(n)$$

Q9

```
int fib(int n)
```

```
{  
    if(n<=1)
```

```
return n;
```

```
return fib(n-1)+fib(n-2);
```

```
}
```

Recurrence:

$$T(n) = T(n-1) + T(n-2)$$

Answer:

$$\Theta(2^n)$$

(More precisely $\Theta(\phi^n)$)

Q10

Arrange in increasing order:

- $n^{3/2}$
- $n \log n$
- $n^{\log_2 n}$
- 2^n

Comparison

$$n \log n < n^{3/2} < n^{\log_2 n} < 2^n$$

Answer:

$$n \log n < n^{3/2} < n^{\log_2 n} < 2^n$$

Q11

$$T(n) = T(n-1) + 1$$

Solution

$$T(n) = n$$

Answer:

$$\Theta(n)$$

Q12

$$T(n)=T(n-1)+n$$

Solution

$$1+2+3+\dots+n$$

Answer:

$$\Theta(n^2)$$

Q13

$$T(n)=2T(n-1)+1$$

Solution

Expands exponentially.

Answer:

$$\Theta(2^n)$$

Q14

$$T(n)=T(n/2)+1$$

Solution

Height of recursion tree:

$$\log n$$

Answer:

$$\Theta(\log n)$$

Q15

```
for(int i=1;i<=n;i*=2)
{
    for(int j=1;j<=n;j++)
    {
    }
}
```

Outer loop:
 $\Theta(\log n)$

Inner loop:
 $\Theta(n)$

Total:

Answer:

$\Theta(n \log n)$

Final Answers (One-Line)

Question	Answer
Q1	$\Theta(n)$
Q2	$a=b$ and $f(n)=\Theta(n)$
Q3	$\Theta(n)$
Q4	$\Theta(n')$

Q5 $\Theta(n \log \log n)$

Q6 $\Theta(n)$

Q7 $\Theta(n^2)$

Q8 $\Theta(n)$

Q9 $\Theta(2^n)$

Q10 $n \log n < n^{3/2} < n^{\log_2 n} < 2^n$

Q11 $\Theta(n)$

Q12 $\Theta(n^2)$

Q13 $\Theta(2^n)$

Q14 $\Theta(\log n)$

Q15

$\Theta(n \log n)$